

READING, PA, USA

WMO: 725103

Lat: 40.373N

Lon: 75.959W

Elev: 105

StdP: 100.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14712

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.2	-20.6	0.6	-11.1	-18.0	0.8	-8.6	12.1	-0.7	11.0	-1.5	4.1	290	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.5	23.7	32.0	23.1	30.6	22.4	25.3	30.8	24.4	29.4	23.7	28.3	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	18.7	27.7	22.9	17.8	26.8	22.3	17.2	26.3	77.6	30.8	74.0	29.5	71.2	28.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.7	7.8	DB	-14.9	36.3	2.8	3.2	-16.9	38.6	-18.5	40.5	-20.1	42.3	-22.1	44.7	(4)
(5)			WB	-15.7	26.8	2.7	1.2	-17.6	27.7	-19.2	28.4	-20.7	29.1	-22.7	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-0.7	0.4	5.2	11.6	17.0	21.8	24.4	23.3	19.6	13.0	7.0	2.0	(6)
(7)		DBStd	9.83	5.73	5.18	5.18	4.71	4.29	3.42	2.88	2.84	3.73	4.41	4.53	4.81	(7)
(8)		HDD10.0	1188	334	270	167	34	1	0	0	0	0	18	112	252	(8)
(9)		HDD18.3	2847	591	501	408	209	78	9	0	2	29	174	339	506	(9)
(10)		CDD10.0	1966	2	2	17	82	219	355	448	414	287	112	22	5	(10)
(11)		CDD18.3	582	0	0	1	7	38	114	189	157	66	9	0	0	(11)
(12)		CDH23.3	5198	0	1	7	87	384	1017	1817	1315	505	64	1	0	(12)
(13)		CDH26.7	1827	0	0	1	23	123	338	738	449	145	10	0	0	(13)
(14)	Wind	WSAvg	2.9	3.4	3.5	3.8	3.5	2.7	2.5	2.3	2.0	2.2	2.5	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	1085	82	67	83	90	100	104	108	88	105	80	88	84	(15)
(16)		PrecMax	1536	250	143	166	201	296	383	296	199	284	271	202	216	(16)
(17)		PrecMin	703	12	16	14	20	22	21	13	4	7	4	13	20	(17)
(18)		PrecStd	189	51	30	35	46	51	66	59	50	66	55	45	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	18.1	24.6	29.4	32.3	34.1	36.9	34.7	32.8	28.0	21.9	18.6	(19)
(20)			MCWB	15.8	13.4	15.9	18.5	22.2	23.9	24.5	23.8	23.0	20.6	16.5	15.5	(20)
(21)		2%	DB	13.7	14.2	19.7	26.1	30.0	32.0	34.0	32.4	30.2	25.2	19.1	14.9	(21)
(22)			MCWB	11.5	9.0	12.9	16.9	21.1	23.3	24.0	23.3	21.7	19.5	14.9	12.5	(22)
(23)		5%	DB	9.7	10.9	16.6	22.9	27.7	30.3	32.3	30.7	28.0	22.8	17.2	12.0	(23)
(24)			MCWB	7.3	6.7	11.1	15.5	19.8	22.2	23.5	22.6	20.8	18.1	13.3	9.6	(24)
(25)		10%	DB	7.1	7.9	13.7	20.3	25.2	28.6	30.7	29.1	26.3	20.9	14.6	9.1	(25)
(26)			MCWB	4.9	4.7	8.9	13.6	18.2	21.1	22.7	21.9	19.8	16.5	11.1	6.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	15.0	16.7	20.5	23.8	25.1	27.1	26.2	24.8	22.9	18.0	15.7	(27)
(28)			MCD B	18.3	17.3	22.0	26.8	30.0	31.3	32.9	32.2	29.5	25.2	20.2	17.6	(28)
(29)		2%	WB	12.0	10.4	14.0	18.6	22.3	24.2	25.6	25.0	23.5	20.9	16.1	12.8	(29)
(30)			MCD B	13.5	13.5	18.1	23.9	28.1	29.8	31.6	29.8	27.3	24.0	18.3	14.8	(30)
(31)		5%	WB	7.8	7.3	11.7	16.6	21.0	23.4	24.7	24.0	22.5	19.1	13.8	10.0	(31)
(32)			MCD B	9.4	9.9	15.5	21.5	25.9	28.5	29.9	28.2	26.0	22.2	16.4	11.5	(32)
(33)		10%	WB	5.2	5.0	9.1	14.8	19.6	22.5	23.8	23.2	21.5	17.1	11.8	6.9	(33)
(34)			MCD B	7.0	8.0	13.1	19.4	23.8	27.1	28.5	27.1	24.8	19.8	14.1	8.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	8.9	9.9	11.3	11.1	10.6	10.8	10.3	10.6	10.4	9.5	7.7	(35)
(36)			MCDBR	11.8	12.9	14.5	15.4	14.1	12.5	12.6	11.8	12.6	12.9	12.8	11.1	(36)
(37)			MCWBR	9.9	9.4	9.5	8.4	6.7	5.3	4.6	4.7	5.6	7.3	9.0	9.3	(37)
(38)		5% WB	MCD BR	11.1	11.6	13.1	13.1	12.3	10.9	10.7	9.8	9.8	10.5	10.5	9.8	(38)
(39)			MCWBR	10.0	9.4	9.8	8.3	6.5	5.3	4.7	4.6	5.3	6.9	8.7	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.327	0.360	0.393	0.433	0.498	0.498	0.497	0.411	0.364	0.326	0.307	(40)	
(41)		taud	2.436	2.359	2.354	2.287	2.209	2.064	2.079	2.069	2.280	2.421	2.492	2.485	(41)	
(42)		Ebn at Noon	852	884	891	880	848	791	787	775	826	832	827	827	(42)	
(43)		Edh at Noon	80	99	112	128	142	164	160	158	120	93	76	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.82	2.73	3.63	4.71	5.27	5.79	5.92	5.14	4.23	2.98	2.09	1.55	(44)	
(45)		RadStd	0.16	0.33	0.28	0.45	0.63	0.48	0.30	0.31	0.54	0.33	0.20	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-120	+172	N/A		
(47) Regional (70 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+74		

Nomenclature: See separate page